

JEE MAIN REVISION MASTER: DIFFERENTIAL EQUATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Linear Differential Equations & Integrating Factor	20 PYQs	Linear Equations
Moderate	Order, Degree & Variable Separable Method	7 PYQs	Differential Equations

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Linear Differential Equations & Integrating Factor	$\text{I.F.} = e^{\int P \, dx}$, $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.} \, dx + C$, $e^{\ln f(x)} = f(x)$
Order, Degree & Variable Separable Method	$\text{Order: Highest derivative order}$, $\text{Degree: Power of highest derivative when polynomial in derivatives}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Linear Differential Equations & Integrating Factor (Very High Yield — 20 PYQs)

Core Principle: First-order Linear Differential Equation in y : $(dy/dx) + P(x)y = Q(x)$. Integrating Factor $\text{I.F.} = e^{\int P \, dx}$.

General solution: $y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) \, dx + C$. If linear in x : $(dx/dy) + P(y)x = Q(y) \implies \text{I.F.} = e^{\int P \, dy}$, solution $x \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) \, dy + C$.

■ **Caution / JEE Pitfall:** Watch for the leading coefficient of (dy/dx) : If written as $x(dy/dx) + y = Q$, divide through by x first to put it in standard form $(dy/dx) + (1/x)y = (Q/x)$!

■ **Speed Trick:** Remember the log-exponential simplification: $\text{I.F.} = e^{\int (2/x) \, dx} = e^{2\ln x} = e^{\ln(x^2)} = x^2$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Sum of the squares of the order and degree (if defined) of a differential equation $2y'' + (y')^2 = \sqrt{(y'')^2 - 3}$ is..." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "Integrating factor of the differential equation $(1+x^2)(dy/dx) + xy = 1$ is..." [Ans: D]
- [2025 Shift] Q#1 (MCQ): "If 'a' and 'b' are the order and degree respectively of the differentiable equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 3 = 0$..." [Ans: N/A]

3.2 Order, Degree & Variable Separable Method (Moderate Yield — 7 PYQs)

Core Principle: Order: highest order derivative occurring in the equation. Degree: power of the highest order derivative when the equation is expressed as a polynomial in derivatives (if $\sin(dy/dx)$ or $e^{dy/dx}$ is present, degree is NOT defined).
Method of Separation of Variables: $(dy/dx) = f(x)g(y)$ implies $\int (dy/g(y)) = \int f(x)dx + C$.

■■ **Caution / JEE Pitfall:** For an equation like $(d^2y/dx^2) + \sin(dy/dx) = 0$, the ORDER is 2, but DEGREE IS NOT DEFINED!

■ **Speed Trick:** The number of arbitrary constants in the general solution of an n-th order differential equation is n. In a PARTICULAR solution, the number of arbitrary constants is always ZERO.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): " $\int xf(x)dx + (f(x)/2) = 0$, then $f(x)$ is equal to..." [Ans: C]
- [2025 Shift] Q#4 (MCQ): "General solution of the differential equation $(dy/dx) + y \tan x = \sec x$ is..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "The degree of the differential equation $1 + ((dy/dx))^2 + (\frac{d^2y}{dx^2})^2 = \sqrt{3}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Sum of the squares of the order and degree (if defined) of a differential equation $2y'' + (y'')^2 = \sqrt{(y''') - 3}$ is

- (A) 3
- (B) 20 [Correct]
- (C) 8
- (D) 16

Correct Answer: B

Solution Analysis: The given differential equation is $2y'' + (y'')^2 = \sqrt{(y''') - 3}$. The highest order derivative present is y''' , so the order of the differential equation is 3. To find the degree, the equation must be written as a polynomial in derivatives. Since y''' is inside a square root, we first remove the radical by squaring both sides: $(2y'' + (y'')^2)^2 = (y''') - 3$. Expanding, $4(y'')^2 + 4y''(y'')^2 + (y'')^4 = y''' - 3$. Bring all terms to one side: $(y''')^2 + 4y''(y'')^2 + 4(y'')^2 - y''' + 3 = 0$. Now it is a polynomial in derivatives. The highest order deriva

Benchmark Problem 2: KCET 2026 (Q#2)

$\int xf(x)dx + (f(x)/2) = 0$, then $f(x)$ is equal to

- (A) e^{-2x}
- (B) e^{2x}
- (C) e^{-x^2} [Correct]
- (D) e^{x^2}

Correct Answer: C

Solution Analysis: Given, $\int xf(x)dx + (f(x)/2) = 0$. Differentiate both sides with respect to x . The derivative of $\int xf(x)dx$ is $xf(x)$, and the derivative of $(f(x)/2)$ is $\frac{f'(x)}{2}$. So, $xf(x) + \frac{f'(x)}{2} = 0$. Multiply by 2: $2xf(x) + f'(x) = 0$. So, $f'(x) = -2xf(x)$. Now separate the variables: $\frac{f'(x)}{f(x)} = -2x$. This means $(1/f(x))(df/dx) = -2x$. Integrate both sides: $\int (1/f)df = \int -2xdx$. $\ln|f| = -x^2 + C$. So, $f(x) = Ce^{-x^2}$. Among the given options, the matching function is e^{-x^2} . So the correct option is Option C.